



Contextual Time Series Embedding: A State Space Perspective for Financial Data

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Abstract

In this work, we propose a novel and principled framework for learning contextual embeddings of financial time series that captures how individual series co-evolve with others over time. Our method combines state space reconstruction with Bayesian nonparametric method to produce embeddings that captures inter-series contextual relationships inferred from intrinsic dynamics. This approach provides a unified theoretical foundation that generalizes several prior methods, while naturally handling missing samples and enabling principled uncertainty quantification.

The proposed approach is validated on two important downstream tasks in finance: Global Industry Classification Standard (GICS) sector classification and portfolio construction. In sector classification, our embeddings consistently outperform recent benchmarks. For portfolio construction, contextual similarity baskets formed using our embeddings achieve superior risk-adjusted returns and enhanced diversification in rolling backtests, highlighting their practical utility. These results demonstrate that the contextual embeddings produced by our framework capture richer and more informative representations of temporal dynamics and context-dependent relationships across time series, enabling more effective and robust financial decision-making.

Keywords

contextual time series embeddings, state space reconstruction, Gaussian process latent variable model, Bayesian nonparametric, GICS sector classification, risk-return portfolio construction

ACM Reference Format:

Guanchao Feng, James Lavinder, Yanfei Huang, Ding Zhao, Morgan Hawkins, Dipanker Koirala, and Dhagash Mehta. 2025. Contextual Time Series Embedding: A State Space Perspective for Financial Data. In *6th ACM International Conference on AI in Finance (ICAIF '25)*, November 15–18, 2025, Singapore, Singapore. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3768292.3771255>

1 Introduction

Financial markets are complex, adaptive systems characterized by stochastic dynamics, structural non-stationarity, and high levels of noise [4, 30]. Time series data produced by these markets, such as asset prices, returns, and volatility, fluctuate at high frequency under the influence of macroeconomic factors, investor behavior, and external shocks [1, 14]. Beyond individual asset fluctuations, market-wide dynamics evolve over time through co-movements, latent dependencies, and regime shifts, posing significant challenges for modeling, forecasting, and learning representations from financial time series [3].

Representation learning has recently emerged as a powerful approach for mapping raw time series into low-dimensional embeddings that capture essential temporal structure [22]. These embeddings help mitigate noise and support tasks such as classification, clustering, anomaly detection, and portfolio optimization [34]. When well-designed, they also enable meaningful similarity comparisons between financial instruments, supporting case-based reasoning and relational inference [13, 24]. Furthermore, time series embeddings further enable multimodal modeling—critical in finance—by bridging sequential signals (e.g., returns, volumes) with static or slow-moving attributes such as sector, size, or fundamentals [9, 23]. Fixed-dimensional representations facilitate integration across modalities in architectures like graph neural networks, transformers, and attention-based systems [18, 35].

Traditional modeling in finance, including autoregressive integrated moving average (ARIMA) models and handcrafted feature engineering, often struggle with high variance, limited generalization, and sensitivity to structural breaks. Deep learning alternatives,

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ICAIF '25, Singapore, Singapore

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ACM ISBN 979-8-4007-2220-2/25/11
<https://doi.org/10.1145/3768292.3771255>

such as recurrent neural networks (RNNs), temporal convolutional networks (TCNs), and transformer-based encoders, have been applied to learn embeddings using objectives like forecasting, reconstruction, or contrastive learning [36, 38]. However, these models typically process time series in isolation and often fail to incorporate relational dynamics across multiple assets. These challenges motivate the need for models that embed each time series in the context of others, capturing dynamic relational information rather than relying on isolated representations [13, 22, 23].

To that end, in this paper we focus on a class of methods that learn contextual, or relational, time series embeddings—that is, representations that capture how each series co-evolves with others and preserve inter-series context for downstream tasks such as sector classification and risk–return portfolio construction. Such approaches have been increasingly adopted in recent literature. For instance, Franceschi et al. [22] and Yue et al. [36] propose unsupervised embedding models that capture inter-series relationships using context-aware objectives. In the financial domain, Dolphin et al. [13] develop a contrastive learning framework where asset embeddings are learned by comparing sub-sequences across time and assets. These methods encode inter-series context via similarity relationships, either implicitly or explicitly, allowing the learned representations to reflect both temporal dynamics and relational structure across series. We make the novel observation that many of these methods are implicitly and closely connected with state space reconstruction (SSR), capturing the latent geometry of underlying system.

Building on this insight, we make the following key contributions: (1) we offer a theoretical foundation that unifies and interprets several prior contextual embedding approaches through the lens of dynamical system theory; and (2) we introduce a novel and principled framework that quantifies inter-series similarity by comparing reconstructed state vectors over time, and integrates this relational (or contextual) information into a Bayesian nonparametric learning framework for embedding, resulting in representations that more faithfully preserve both temporal dynamics and relational structure. This dynamical perspective not only grounds the proposed approach of contextual embeddings in a solid theoretical framework, but also enables improved performance on downstream financial applications. We validate the effectiveness of our learned embeddings using two important financial applications with an real-world dataset: (1) Global Industry Classification Standard (GICS) sector classification, which assesses whether the embeddings capture economically meaningful structure across equities, and (2) risk–return portfolio construction, which evaluates the utility of the embeddings for building diversified equity portfolios (or equity baskets) with improved risk-adjusted returns.

2 Background

2.1 SSR and Takens’ Theorem

To capture the underlying dynamics of a time series, we base our methodology on the theory of dynamical systems. In particular, we employ SSR, also known as phase space reconstruction, grounded in Takens’ theorem [32], a cornerstone result in nonlinear time series analysis and dynamical systems theory.

THEOREM 1 (TAKENS’ THEOREM). *Let $\mathcal{M}$ be a compact manifold of (integer) dimension d . Then for generic pairs (ϕ, y) , where*

- $\phi: \mathcal{M} \rightarrow \mathcal{M}$ is a C^2 -diffeomorphism of $\mathcal{M}$ in itself,
- $y: \mathcal{M} \rightarrow \mathbb{R}$ is a C^2 -differentiable function,

the map $\Phi_{(\phi, y)}: \mathcal{M} \rightarrow \mathbb{R}^{2d+1}$ given by

$$\Phi_{(\phi, y)}(x) := (y(x), y(\phi(x)), y(\phi^2(x)), \dots, y(\phi^{2d}(x)))$$

is an embedding of $\mathcal{M}$ in $\mathbb{R}^{2d+1}$.

In practice, the delay coordinate map (or delay embedding) is the most widely used instantiation of the map ϕ . Takens’ theorem states that, under generic conditions, the state space of a dynamical system can be reconstructed from time-delayed observations of a single scalar variable. Specifically, given a time series $\{x_t^{(i)}\}_{t=1}^T$, delay embedding constructs a sequence of state vectors in an E -dimensional space to reconstruct the dynamics of the underlying system. The delay-embedded state vector at time t is defined as

$$\mathbf{m}_x[t] = [x_t, x_{t-\tau}, x_{t-2\tau}, \dots, x_{t-(E-1)\tau}], \quad (1)$$

where τ is the *time delay* and E is the *embedding dimension*. The reconstructed state space, or, the *shadow manifold*, is then given by

$$\mathbf{M}_x = \{\mathbf{m}_x[t] \in \mathbb{R}^E \mid t = (E-1)\tau + 1, \dots, T\}. \quad (2)$$

Under Takens’ embedding theorem, if the original dynamical system possesses an attractor of dimension d , then a delay embedding of dimension $E \geq 2d + 1$ is sufficient to reconstruct the system’s state space. Specifically, the reconstructed manifold $\mathbf{M}_x$ is diffeomorphic to the original attractor $\mathcal{M}$, meaning that it preserves the topological and dynamical structure of the system up to a smooth coordinate transformation.

2.2 Related Work

A growing class of methods—termed *contextual time series embeddings*—aims to learn representations shaped not only by intra-series dynamics but also by relationships to other series in the dataset. These methods define inter-series similarity—via alignment, predictive influence, or learned kernels—and use it to derive low-dimensional embeddings that capture both temporal structure and relational context. For example, a method proposed in [27] constructs a pairwise similarity matrix using dynamic time warping (DTW) and applies low-rank matrix factorization to obtain embeddings that preserve DTW-induced relational structure.

Several recent approaches for financial time series can be interpreted as performing a form of SSR: each time series is mapped to a sequence of state vectors, and pairwise similarities between these vectors are used to compute embeddings. In [11], fixed-length sub-sequences are extracted as

$$c(i, t, n_s) = [x_{t-n_s+1}^i, x_{t-n_s+2}^i, \dots, x_t^i], \quad (3)$$

and a similarity matrix is built by counting how often $c(i, t, n_s)$ appears among the k -nearest neighbors of sub-sequences from other series. Embeddings are then obtained via low-rank matrix factorization with ℓ_2 regularization. Comparing Equation (3) to the delay embedding in Equation (1), each sub-sequence corresponds to a reconstructed state vector with embedding dimension $E = n_s$ and lag $\tau = 1$. A similar interpretation applies to [12], where embeddings are learned by training a neural network to predict each stock’s

top- k most similar context stocks, using only time periods with large return magnitudes. This setup can be viewed as comparing state vectors sampled from shadow manifolds reconstructed with embedding dimension $E = 1$ and lag $\tau = 1$. In [13], a co-occurrence matrix C is constructed to extract embeddings, where each entry C_{ij} counts how often asset α_j appears among the top- k most similar assets to α_i within sliding windows of length 22 and stride 5. This procedure is equivalent to constructing delay embeddings with $E = 22$ and $\tau = 1$, followed by uniform downsampling every 5 steps.

In contrast to prior approaches, our method provides a more principled and robust framework for contextual time series embedding. First, while earlier methods often perform a form of SSR implicitly, they typically fix critical parameters—such as the embedding dimension (E) and lag (τ)—empirically, with little theoretical justification or sensitivity analysis. Second, the construction of similarity matrices in these methods is typically ad hoc: similarity is aggregated by counting co-occurrences within the top- k nearest neighbors, where k is a key hyperparameter lacking principled guidance for selection. In our approach, we instead model the full distribution of similarity values across time, enabling a more stable and interpretable representation of inter-series relationships.

Moreover, unlike methods that rely on linear approximations such as low-rank matrix factorization, we cast the embedding problem as Bayesian inference under Gaussian Processes (GPs) framework which has been successfully adopted for various machine learning tasks [6]. This allows for flexible nonlinear modeling and principled uncertainty quantification, yielding embeddings that capture rich, nonlinear structure while remaining robust to noise, missing data, and other irregularities common in financial time series. Finally, our method naturally accommodates multivariate time series and missing samples, making it broadly applicable in real-world financial settings.

2.3 Evaluation Tasks and Practical Utility

2.3.1 Task 1: GICS Sector Classification. Sector classification is a foundational task in financial analysis with broad practical relevance. The GICS groups companies into sectors based on their primary lines of business. For instance, sector-level exposures underpin common factor models like the Fama–French industry factors [17] and guide asset allocation decisions. In practice, sector labels are not always available or reliable, for example, for newly listed securities, foreign or dual listed securities, or proprietary instruments. In such cases, inferring sector membership directly from return dynamics can offer a data-driven alternative to fundamental or text-based approaches.

Sector classification has also become a widely used benchmark for evaluating time series embeddings in finance [11–13]. It serves as a practical downstream task for assessing whether embeddings capture economically meaningful patterns. High classification accuracy implies that the learned representations preserve latent sector structure, such as co-movement patterns, volatility profiles, and systematic responses to market conditions. This task provides a natural and interpretable testbed for validating learned embeddings, particularly in terms of their ability to recover economic labels from raw financial time series.

2.3.2 Task 2: Risk-Return Basket Construction. In equity portfolio (or basket) construction, the relationship between the risk of a single security and that of other assets is fundamental to achieving effective risk-adjusted returns. A key aspect of this relationship is the correlation of returns between an individual equity and a diversified portfolio. While modern portfolio theory demonstrates that diversification reduces overall portfolio risk [15], understanding diversification in the presence of correlated assets, especially those with shared return dynamics, requires deeper exploration.

In this task, we investigate whether our proposed contextual return time series embeddings can support this process. For each target security α_i , we construct a contextual similarity basket $\mathcal{P}_{\alpha_i}$ by selecting securities with high cosine similarity in the learned embedding space. The goal is to assess whether $\mathcal{P}_{\alpha_i}$ can improve the risk-return trade-off compared to holding α_i alone. Our primary hypothesis is that embeddings derived from net daily returns can capture economically meaningful relationships between securities. Building on this, we propose an additional hypothesis: that such embeddings can inform portfolio construction in a way that yields more effective risk-adjusted diversification, even when centered around a single target security α_i .

A key theoretical motivation for this hypothesis lies in the observation that return co-movements across securities are often driven by latent macroeconomic and global factors—not solely by shared industry affiliation. For instance, Metaplanet, a Japan-based firm pursuing a Bitcoin treasury strategy, is classified under the Hotels sector, whereas MicroStrategy, a U.S.-based company with a similar Bitcoin-focused balance sheet, remains categorized as a Software firm. Despite their differing sector labels, the two firms exhibit highly similar return behavior driven by their shared exposure to Bitcoin. As a result, stocks from different sectors may exhibit stronger return similarity with a given target security than others within the same sector, some examples, found in the dataset we used for experiments, are illustrated in Figure 2 and discussed in Section 4.1.2. These relationships can stem from common exposures to inflation, interest rate regimes, geopolitical risks, or macroeconomic trends. Embedding-based similarity can therefore uncover cross-sector return dynamics that traditional sector-based classifications overlook, potentially enhancing diversification and improving portfolio efficiency [2, 16].

3 Methodology

In this section, we present our proposed time series embedding learning framework in detail. We begin by formally defining the problem setting. Let $\mathbf{D} \in \mathbb{R}^{N \times T}$ denote a dataset consisting of N time series, each of length T . The i -th row of $\mathbf{D}$, denoted $\mathbf{D}_{i,:}$, corresponds to the time series $\{x_t^{(i)}\}_{t=1}^T$. Without loss of generality, we assume that each $\{x_t^{(i)}\}_{t=1}^T$ is a regularly sampled univariate time series. Multivariate time series can be readily accommodated within our framework. Our goal is to construct a low-dimensional representation of $\mathbf{D}$, denoted $\tilde{\mathbf{D}} \in \mathbb{R}^{N \times Q}$, where the i -th row corresponds to an embedding of the time series $\{x_t^{(i)}\}_{t=1}^T$ in a Q -dimensional latent space, with $Q \ll T$. The representation $\tilde{\mathbf{D}}$ is designed to preserve similarities among time series in $\mathbf{D}$ across time and to support efficient modeling in downstream machine learning tasks.

The proposed methodology consists of three components, which we describe in detail in the following subsections.

3.1 State Space Reconstruction

Since our notion of time series similarity is grounded in dynamical systems theory, the first step of the proposed approach is to reconstruct the state space for each time series in $\mathbf{D}$ using Equations (1) and (2). Although Takens' theorem guarantees that an embedding dimension of $E = 2d + 1$ suffices to reconstruct a d -dimensional manifold, practical implementation poses challenges: the true attractor $\mathcal{M}$ is latent, and its intrinsic dimension d is unknown. Similarly, the delay parameter τ is unconstrained in theory, but its choice greatly influences the quality of reconstruction in finite-length time series. Too small a τ yields redundant coordinates, while too large a τ can lead to loss of dynamical information [5, 8, 21].

To select suitable delay embedding parameters for SSR, we adopt two well-established heuristics:

- The time delay τ is estimated using *average mutual information* (AMI) [25], which selects the first local minimum of the mutual information between $\{x_t\}$ and $\{x_{t+\tau}\}$, ensuring each embedding coordinate adds non-redundant information.
- The embedding dimension E is chosen via the *false nearest neighbors* (FNN) method [28], which identifies the smallest dimension where spurious neighbors due to projection are minimized, preserving the topology of the underlying attractor.

In this step, for each time series $\{x_t^{(i)}\}_{t=1}^T$ in the dataset $\mathbf{D}$, we estimate the embedding dimension E_i and time delay τ_i using the heuristics described above. To construct the shadow manifolds $\{\mathbf{M}_{x_i}\}_{i=1}^N$, we define global embedding parameters by taking the maximum embedding dimension and the minimum time delay across all series: $\tilde{E} = \max(\{E_i\}_{i=1}^N)$ and $\tilde{\tau} = \min(\{\tau_i\}_{i=1}^N)$. This choice ensures that the reconstructed state space is sufficiently expressive to capture the dynamics of each time series while maintaining temporal resolution. For instance, Figure 1 shows the distributions of $\{E_i\}_{i=1}^N$ and $\{\tau_i\}_{i=1}^N$ computed from a dataset, contains return series of 611 company stocks, that is adopted in our experiments (see Section 4). It can be observed in Figure 1 that for this dataset, $\tilde{E} = 5$ and $\tilde{\tau} = 1$.

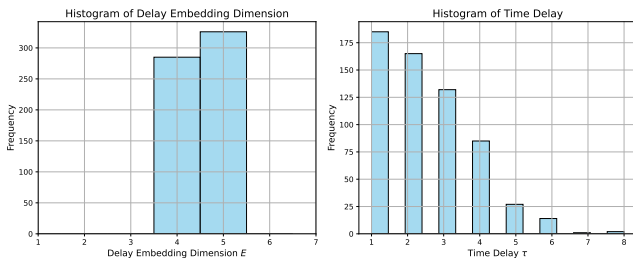


Figure 1: Histograms of $\{E_i\}_{i=1}^N$ (left) and $\{\tau_i\}_{i=1}^N$ (right) derived from the company stocks dataset adopted for experiments described in Section 4.

3.2 Similarity Matrix Construction

Due to nonstationarities, dynamical complexity, and heteroscedastic noise commonly present in financial time series, direct comparison of raw observations can be unreliable for quantifying similarity. Instead, we anchor the notion of similarity on the state vectors reconstructed in the shadow manifolds, which more faithfully capture the underlying system dynamics. To be more specific, let $\text{sim}(\cdot, \cdot)$ denote a similarity function $\text{sim} : \mathbb{R}^E \times \mathbb{R}^E \rightarrow \mathbb{R}$. For each time instant t , we construct a $N \times N$ similarity matrix $\mathbf{S}_{:,t} := \mathbf{S}[:, :, t] \in \mathbb{R}^{N \times N}$, where the entry at the i -th row and j -th column is computed as:

$$S[i, j, t] = \text{sim}(\mathbf{m}_{x_i}[t], \mathbf{m}_{x_j}[t]), \quad (4)$$

where $\mathbf{m}_{x_i}[t] \in \mathbb{R}^E$ and $\mathbf{m}_{x_j}[t] \in \mathbb{R}^E$ are state vectors at time instant t from the reconstructed shadow manifolds $\mathbf{M}_{x_i}$ and $\mathbf{M}_{x_j}$, respectively. Essentially, $\mathbf{S} \in \mathbb{R}^{N \times N \times [T - (E-1)\tau]}$, and the vector $\mathbf{S}_{i,j,:} := \mathbf{S}[i, j, :] \in \mathbb{R}^{[T - (E-1)\tau]}$ captures the time-varying similarity between time series i and j in the dataset $\mathbf{D}$. In this work, we use cosine similarity when measuring similarity between state vectors.

In order to properly summarize the similarity between time series i and j over time, we first apply min-max normalization to their historical similarity measurements at each time point:

$$\mathbf{S}^{\text{norm}}[i, j, t] = \frac{S[i, j, t] - \min_t S[i, j, t]}{\max_t S[i, j, t] - \min_t S[i, j, t]}, \quad (5)$$

where $\max_t S[i, j, t]$ and $\min_t S[i, j, t]$ denote the maximum and minimum similarity values observed over time between the state vectors of time series i and j . Then, we average these normalized similarity values over the set of time points $\mathcal{T}_{i,j}$ where the similarity is observed:

$$\tilde{\mathbf{S}}[i, j] = \frac{1}{|\mathcal{T}_{i,j}|} \sum_{t \in \mathcal{T}_{i,j}} \mathbf{S}^{\text{norm}}[i, j, t], \quad (6)$$

with $\mathcal{T}_{i,j} = \{t : \mathbf{S}^{\text{norm}}[i, j, t] \text{ is not missing}\}$. In other words, the similarity between time series i and j is computed by first normalizing all available historical similarity measurements to the $[0, 1]$ range as shown in Equation (5), then averaging these normalized values over time as suggested in Equation (6), yielding a single aggregated similarity score $\tilde{\mathbf{S}}[i, j]$ that captures their overall dynamic similarity.

3.3 Embedding Extraction

The similarity matrix $\tilde{\mathbf{S}} \in \mathbb{R}^{N \times N}$ summarizes the similarities between the time series in the dataset $\mathbf{D}$. We adopt $\tilde{\mathbf{S}}$ to infer $\tilde{\mathbf{D}}$, the low-dimensional representation of the original dataset $\mathbf{D}$. Formally, $\tilde{\mathbf{D}} = [\tilde{\mathbf{d}}_1^T \ \dots \ \tilde{\mathbf{d}}_N^T]^T \in \mathbb{R}^{N \times Q}$, where each row $\tilde{\mathbf{d}}_i \in \mathbb{R}^Q$ is the latent embedding associated with the i -th time series (i.e., row) in $\mathbf{D}$. To this end, we employ the Gaussian process latent variable model (GPLVM) [26, 33], which provides a flexible nonlinear dimensionality reduction approach within a Bayesian probabilistic framework.

We assume the following generative process for the similarity matrix $\tilde{\mathbf{S}} = f(\tilde{\mathbf{D}}) + \epsilon$, where $f : \mathbb{R}^{N \times Q} \rightarrow \mathbb{R}^{N \times N}$ is an unknown vector-valued function, and $\epsilon \in \mathbb{R}^{N \times N}$ is a noise matrix whose rows are independent, zero-mean Gaussian vectors with covariance $\sigma_\epsilon^2 \mathbf{I}$. Conceptually, we posit that the observed similarity matrix $\tilde{\mathbf{S}}$

arises from a nonlinear mapping of low-dimensional latent coordinates $\tilde{\mathbf{D}} \in \mathbb{R}^{N \times Q}$, which represent a low-dimensional manifold underlying the data, and our goal is to learn $\tilde{\mathbf{D}}$ under some assumptions of $\tilde{\mathbf{f}}$. Specifically, the j -th column of the similarity matrix, denoted $\tilde{\mathbf{S}}_{:,j} \in \mathbb{R}^N$, is generated as

$$\tilde{\mathbf{S}}_{:,j} = f_j(\tilde{\mathbf{D}}) + \epsilon_j, \quad j = 1, \dots, N,$$

where each latent function $f_j : \mathbb{R}^{N \times Q} \rightarrow \mathbb{R}^N$ is drawn independently from a Gaussian process prior, i.e., $f_j \sim \mathcal{GP}(0, k(\cdot, \cdot))$, and $\epsilon_j \sim \mathcal{N}(0, \sigma_\epsilon^2 \mathbf{I})$ represents independent Gaussian noise. Conditioned on the $\tilde{\mathbf{D}}$, each column of $\tilde{\mathbf{S}}$ is therefore modeled as a realization from $\mathcal{GP}(0, k(\cdot, \cdot))$. This conditional independence across columns yields a factorized likelihood

$$p(\tilde{\mathbf{S}} | \tilde{\mathbf{D}}, \theta) = \prod_{j=1}^N \mathcal{N}(\tilde{\mathbf{S}}_{:,j} | \mathbf{0}, \mathbf{K} + \sigma_\epsilon^2 \mathbf{I}),$$

where $\mathbf{K} \in \mathbb{R}^{N \times N}$ is the covariance matrix with entries $K_{ij} = k(\tilde{\mathbf{d}}_i, \tilde{\mathbf{d}}_j)$, and θ collects the hyper-parameters of $\mathcal{GP}$ including the noise variance σ_ϵ^2 .

A standard Gaussian prior is placed over the latent embeddings, i.e., we initialize $p(\tilde{\mathbf{D}}) = \prod_{i=1}^N \mathcal{N}(\tilde{\mathbf{d}}_i | \mathbf{0}, \mathbf{I})$. Due to the nonlinear mapping, the posterior $p(\tilde{\mathbf{D}} | \tilde{\mathbf{S}})$ is intractable. We use the variational framework proposed in [33], where $p(\tilde{\mathbf{D}} | \tilde{\mathbf{S}})$ is approximated with a Gaussian variational distribution $q(\tilde{\mathbf{D}})$ by maximizing the evidence lower bound (ELBO):

$$\mathcal{L} = \mathbb{E}_{q(\tilde{\mathbf{D}})} \left[\log p(\tilde{\mathbf{S}} | \tilde{\mathbf{D}}, \theta) \right] - \text{KL}(q(\tilde{\mathbf{D}}) \| p(\tilde{\mathbf{D}})).$$

The mean of $q(\tilde{\mathbf{D}})$ is adopted as the embedding $\tilde{\mathbf{D}}$, and the covariance matrix of $q(\tilde{\mathbf{D}})$ encodes the uncertainty in embedding learning.

The design of covariance function $k(\cdot, \cdot)$ encodes our prior knowledge and assumptions of the unknown mappings [20]. The covariance function utilized in this work is shown in Equation (7) which is a radial basis function (RBF):

$$k(\tilde{\mathbf{d}}_i, \tilde{\mathbf{d}}_j) = \sigma_f^2 \exp \left(-\frac{1}{2} \sum_{q=1}^Q \frac{(\tilde{d}_{i,q} - \tilde{d}_{j,q})^2}{\ell_q^2} \right). \quad (7)$$

This covariance function encodes our assumption that the unknown mapping $\{f_j\}_{i=1}^N$ are smooth since RBF is infinitely differentiable. Additionally, this covariance function assigns a length scale ℓ_q^2 for each latent dimension q to measure its importance or relevance of q -th dimension, which will be learned from data through the optimization. This is known as automatic relevance determination (ARD) which makes covariance function more expressive. A higher ARD weight (inverse length-scale, i.e., $1/\ell_q^2$) indicates that the corresponding q -th latent dimension plays a more significant role in explaining the similarity structure of the data. In contrast, dimensions with small ARD weights contribute little and can be interpreted as less informative. Naturally, the notion of signal-to-noise-ratio (SNR) can be carried out by $\sigma_f^2/\sigma_\epsilon^2$ which measures how much of the variation in $\tilde{\mathbf{S}}$ is explained by the latent function $\mathbf{f}$ (i.e., the signal), versus how much is due to observational noise.

3.4 Handling Missing Observations and Multivariate Time Series

Real-world time series can exhibit multivariate structure (e.g., physiological or economic indicators) and may contain missing observations. Our framework is naturally equipped to handle both challenges. First, SSR with delay embeddings has been shown to be robust to missing samples and can be extended to multivariate settings. In fact, multivariate delay embeddings often improve the fidelity of the reconstructed state by leveraging complementary information across variables [10]. Second, at each time point t , the similarity matrix $\mathbf{S}_{:,t}$ can be constructed as long as the similarity between state vectors is well-defined. When state vectors are drawn from multivariate shadow manifolds, standard similarity measures such as cosine similarity remain applicable. Moreover, the framework allows for dimension-specific weighting—enabling reduced influence from dimensions known to be corrupted by noise or unreliable measurements. While missing observations in the time series can result in incomplete similarity matrices $\mathbf{S}_{:,t}$ at specific time steps, our use of temporal summary statistics over $\mathbf{S}_{:,t}$ mitigates the impact of such sparsity, making the approach robust to missing samples. Finally, in the embedding learning step, the GPLVM framework naturally accommodates missing samples by modeling the observed data probabilistically as draws from a multivariate Gaussian conditioned on latent variables. This allows the model to marginalize missing observations in learning, effectively handling incomplete data without requiring imputation or data preprocessing.

4 Experiments and Results

In our experiments, we use a publicly available real-world dataset containing daily returns of 611 publicly traded U.S. companies from 2000 to 2018, along with sector and industry labels based on the GICS [12]. This dataset captures realistic financial dynamics, making it well-suited for evaluating the practical performance of time series embedding methods. In particular, it is the same dataset used in [11–13] to assess return series embeddings through the GICS sector classification task.

To demonstrate the practical utility of our learned time series embeddings, we evaluate them on two downstream tasks: sector classification and risk-return basket construction. The two-task design offers a complementary evaluation: the sector-classification task benchmarks the learned embeddings against established methods [11–13], while the portfolio-construction task examines how the same representations affect risk and diversification outcomes. These tasks are described in detail in the following subsections. For embedding learning using our proposed approach, we set the SSR parameters to $\tilde{E} = 5$ and $\tilde{\tau} = 1$, as illustrated in Figure 1. We primarily use an embedding dimensionality of $Q = 50$, which is larger than those adopted in prior work [11–13] (typically $Q = 15$ to $Q = 20$), in order to improve the SNR during representation learning with GPLVM. Empirically, we observe that the SNR with $Q = 50$ is approximately $1.5\times$ that of $Q = 15$.

To ensure a fair comparison in the sector classification task, we additionally evaluate our method using $Q = 15$. This comparison demonstrates that our performance gains are not solely attributable

to increased embedding dimensionality. For the risk-return basket construction task, we report results using $Q = 50$ only.

Table 1: Sector classification performance with different embeddings and classifiers. Top: Proposed method ($Q=50$ and $Q=15$). Bottom: Baselines from prior work. All scores are averaged over five cross-validation folds.

Embedding / Classifier	Q	F1 (Macro)	F1 (Micro)	F1 (Weighted)
Proposed Method				
Ours + kNN ($k=5$)	50	0.707	0.725	0.723
	15	0.700	0.712	0.711
Ours + Random Forest	50	0.764	0.771	0.769
	15	0.719	0.745	0.735
Ours + SVM	50	0.752	0.774	0.769
	15	0.723	0.746	0.739
Baselines				
Distributed + SVM [12]	20	—	0.600	0.600
Case-based + kNN [11]	15	—	0.640	0.640
Contrastive + SVM [13]	16	—	0.690	0.690

4.1 Task 1: Sector Classification

To evaluate the quality of the learned time series embeddings, we adopted the same classification task and experimental setup as in [11–13]. Specifically, given the historical daily return time series of 611 publicly traded U.S. companies from 2000 to 2018, we learned an embedding for each return time series and used it to predict the corresponding GICS sector label. This multiclass classification problem involved 11 sector classes, characterized by notable class imbalance, with sector sizes ranging from 14 to 88.

4.1.1 Quantitative Evaluation. By closely following the experimental setup of [11–13], we ensured direct comparability with previously reported classification results. Specifically, we adopted the time series embedding methods from these works as baselines and evaluated all models using 5-fold cross-validation, training a separate classifier for each fold to predict the GICS sector label. Final performance was reported as the average of classification metrics across the five folds.

In addition to Support Vector Machines (SVM) and k -Nearest Neighbors (kNN, with $k = 5$ and Euclidean distance), both used in the referenced studies, we also included the Random Forest (RF) classifier, motivated by its strong performance on tabular data reported in the literature [19]. All classifiers were implemented using the `scikit-learn` library, with default hyperparameters. We repeated the experiments three times, and the reported results reflect average performance across all runs.

GICS sector classification serves as a quantitative proxy for assessing the quality of the learned embeddings. Evaluation metrics such as F1 score, computed under macro, micro, and weighted averaging, provide a robust basis for comparison. We summarize sector classification performance in Table 1, including results from our proposed embedding method paired with three standard classifiers, as well as baseline methods reported in prior work. All three classifiers, when combined with our embeddings, outperformed

the strongest baseline reported in Dolphin et al. [13]. Notably, our method surpassed the best baseline even when using a reduced embedding dimensionality of $Q = 15$, suggesting that the gains were not simply due to increased representational capacity. Among the tested classifiers, the RF classifier achieved the best overall performance, yielding the highest F1 (Macro) and F1 (Weighted) scores. These strong classification results suggest that our learned embeddings capture economically meaningful temporal dynamics, including sector-level co-movements, volatility characteristics, and systematic responses to market conditions.

4.1.2 Qualitative Evaluation of Embeddings. Beyond classification accuracy, the learned embeddings also provide qualitative insight into the underlying economic structure. If the embeddings are meaningful, companies within the same sector, subject to similar macroeconomic forces and industry dynamics, should exhibit comparable return behavior over time. This structure should manifest as clustering in the embedding space. Figure 2 illustrates this property by visualizing companies from four major GICS sectors projected onto the latent spaces spanned by the top three and top two dominant latent dimensions, respectively. Dominant dimensions were identified based on ARD weights. As shown, companies from the same sector tended to form coherent clusters, suggesting that the embeddings effectively captured sector-level patterns.

Moreover, the embedding space revealed nuanced relationships that extended beyond formal sector assignments. For instance, **Tide-water, Inc.** (TDW), officially categorized under *Consumer Services*, was embedded near companies from the *Energy* sector—reflecting its actual business operations as a provider of offshore petroleum services. Similarly, **Williams Companies Inc.** (WMB), labeled as *Public Utilities*, clustered alongside *Energy* firms due to its core focus on natural gas processing and transportation. Another comparable case was observed with **Paychex, Inc.** (PAYX), classified under *Consumer Services* but embedded near *Finance* sector companies, reflecting its role in payroll and HR outsourcing—whose operations, revenue model, and macroeconomic sensitivity more closely align with financial services. These examples underscore the ability of the learned embeddings to capture economically meaningful similarities not always reflected in GICS sector labels. Similar observations, described as “high similarity mismatches,” were also reported in Dolphin et al. [13].

4.2 Task 2: Risk-Return Basket Construction

The basket-construction study tests whether the embeddings, previously assessed through sector classification, can also inform risk-return decisions beyond sector heuristics. Specifically, for each security α_i , a contextual similarity basket $\mathcal{P}_{\alpha_i} = \left\{ \alpha_i^{\text{neighbour}_h} \right\}_{h=1}^{H_i}$ is constructed by selecting H_i securities whose embedding vectors exhibit high cosine similarity to that of α_i , excluding α_i itself. To determine H_i , we retain only those stocks whose similarity scores fall within the top decile, i.e.,

$$\text{sim}(\alpha_i, \alpha_j) \geq \text{Percentile}_{90} \left(\{ \text{sim}(\alpha_i, \alpha_k) \}_{k \neq i} \right),$$

where $\text{sim}(\alpha_i, \alpha_j)$ denotes the cosine similarity between the embedding of α_i and that of α_j . The weight of each security in $\mathcal{P}_{\alpha_i}$ is computed by normalizing its similarity to α_i over the sum of similarities of all H_i securities to α_i (in the embedding space).

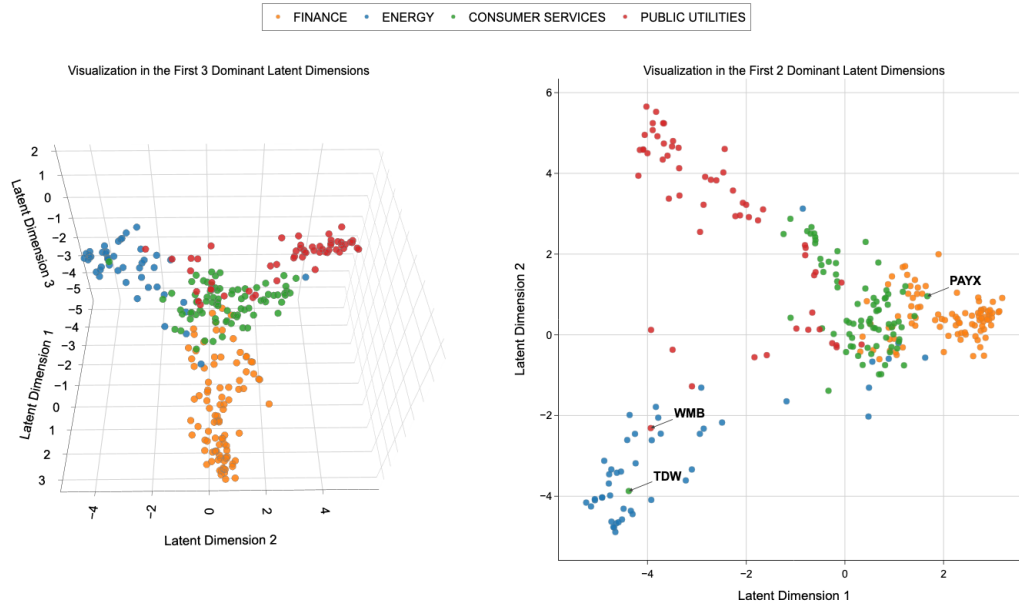


Figure 2: Visualization of learned time series embeddings for companies across four major GICS sectors. Each point represents a company, color-coded by its GICS sector. The left panel shows a 3D projection onto the first three dominant latent dimensions; the right panel shows a 2D projection onto the top two. Clusters largely align with GICS sector labels, indicating that the learned embeddings capture sector-level structure. Notably, several companies are embedded closer to economically similar firms from different GICS sectors, revealing nuanced relationships beyond formal sector assignments.

Given a universe of 611 securities in the dataset, this typically results in portfolios containing approximately 60 stocks. The literature on optimal portfolio size for diversification varies: while early studies suggested that 10 securities may suffice, more recent work recommends holding 30–50 securities, and in some cases up to 100, to achieve meaningful diversification [37]. Our construction method, which yields portfolios based on the 90th percentile similarity threshold, therefore lies within the empirically supported range for effective diversification.

4.2.1 Evaluation Metrics and Baseline Methods. To assess the performance of contextual similarity baskets $\{\mathcal{P}_{\alpha_i}\}_{i=1}^{611}$, we adopt a comprehensive set of evaluation metrics. For each metric, we report the average performance across all 611 baskets, each corresponding to a unique target security α_i .

- **Sharpe Ratio Δ :** Measures the improvement in risk-adjusted return, defined as the difference in Sharpe ratios between $\mathcal{P}_{\alpha_i}$ and α_i , i.e., $\text{Sharpe}_{\mathcal{P}_{\alpha_i}} - \text{Sharpe}_{\alpha_i}$. The Sharpe Ratio is computed as the excess return minus the risk-free rate (set to 0) divided by standard deviation of returns (volatility). Higher Δ values indicate better risk-adjusted performance from embedding-informed diversification [29].
- **Return (Cumulative) Δ :** Captures the change in total return (denoted as R) over the test period between $\mathcal{P}_{\alpha_i}$ and α_i , i.e., $R_{\mathcal{P}_{\alpha_i}} - R_{\alpha_i}$. A positive Δ indicates that the embedding-based basket achieved higher cumulative return than α_i .
- **Sortino Ratio Δ :** Measures improvement in downside risk-adjusted return. The Sortino ratio considers only downside volatility, making it more appropriate when upside variance

is not penalized [31]. A higher Δ reflects better downside risk-adjusted performance.

- **Drawdown Reduction:** Quantifies the reduction in maximum drawdown achieved by the contextual similarity basket. Defined as $1 - \left| \frac{\text{MaxDD}_{\mathcal{P}_{\alpha_i}}}{\text{MaxDD}_{\alpha_i}} \right|$, where MaxDD (stands for maximum drawdown [7]) is the maximum peak-to-trough decline. Higher values imply better downside protection.
- **Correlation (Pearson):** Measures the linear correlation between daily returns of $\mathcal{P}_{\alpha_i}$ and α_i . High correlation suggests that embedding-based portfolios preserve return comovements, indicating that embeddings encode meaningful financial structure.
- **Variance Reduction:** Measures diversification benefits of the contextual similarity basket. Defined as $1 - (\sigma_{\mathcal{P}_{\alpha_i}}^2 / \sigma_{\alpha_i}^2)$, where σ^2 denotes return variance. Higher values reflect greater volatility reduction.

When comparing the performance of a target security α_i to its contextual similarity basket $\mathcal{P}_{\alpha_i}$, there is a risk that improvements in risk-adjusted metrics may arise from reduced volatility alone, rather than from a genuine enhancement of the risk-reward profile. To control for this, we include two baseline portfolio construction strategies for comparison:

- (1) **Correlation-based portfolio construction:** For each target security α_i , we identify the top decile of securities with the highest Pearson correlation over the training period. This portfolio serves as a baseline for traditional return-based similarity, allowing us to benchmark the added value of our contextual similarity embedding-based selection.

Table 2: Performance Metrics for Different Strategies and Configurations.

Strategy/Training Window	Sharpe Ratio Δ	Return Δ	Sortino Ratio Δ	Drawdown Reduction	Correlation	Variance Reduction
Industry-based (10-year)	0.149	0.16%	0.116	0.105	0.610	0.214
Correlation-based (10-year)	0.235	-1.07%	0.185	0.293	0.631	0.484
Contextual similarity-based (10-year)	0.288	0.31%	0.225	0.303	0.651	0.495
Industry-based (3-year)	0.172	0.29%	0.145	0.107	0.593	0.233
Correlation-based (3-year)	0.248	-1.81%	0.204	0.288	0.624	0.477
Contextual similarity-based (3-year)	0.337	0.49%	0.284	0.289	0.638	0.513

- (2) **Industry-based portfolio construction:** This portfolio includes all securities in the same GICS sector or industry group as the target security α_i , excluding α_i itself. Since securities within the same industry typically share common risk factors, this serves as a strong baseline for evaluating whether embeddings capture more informative cross-sector relationships than traditional classifications.

4.2.2 Rolling Backtesting and Results. To capture time-varying market dynamics, we use a rolling backtesting framework with fixed-length training windows followed by one-year test periods. Training entails learning contextual embeddings from the training data, which are then used to construct portfolios for the subsequent test year. Both training and test windows advance by one year at each step. We evaluate two settings: (1) 10 years of training, (2) 3 years of training, each followed by a 1-year test. This approach enables robust out-of-sample evaluation across diverse market regimes. Performance metrics are averaged over all test periods and summarized in Table 2.

The contextual similarity-based strategy consistently outperforms both baseline strategies across all major evaluation metrics under both rolling backtest configurations. Notably, the 3-year training window yields the strongest overall performance, suggesting that shorter-term historical context may be particularly informative for capturing relevant return dynamics. The 3-year contextual strategy achieves a Sharp Ratio Δ of 0.337, compared to 0.172 for the industry-based strategy and 0.248 for the correlation-based strategy. The 10-year contextual strategy yields a Sharp Ratio Δ of 0.288, relative to 0.149 and 0.235 for the respective baselines. Similarly, in terms of cumulative return improvement, the 3-year contextual strategy achieves a relative gain of 0.49%, outperforming the industry-based baseline (0.29%) and the correlation-based baseline (-1.81%). The 10-year strategy also performs strongly, achieving a return delta of 0.31%, versus 0.16% and -1.07% for the industry and correlation strategies, respectively.

These performance gains are consistent across all risk-adjusted metrics. Together, these results highlight that contextual similarity not only enhances diversification but also improves the overall risk-return profile. This supports our central hypothesis that contextual embedding-based similarity captures a richer economic structure than traditional heuristics, yielding a more effective risk-return portfolio construction.

5 Conclusion

We introduced a principled and robust framework for learning contextual embeddings of financial time series from a state space

perspective. Unlike traditional methods that treat each series in isolation, our approach defines similarity in a context-aware fashion, capturing how time series co-evolve over time. This yields embeddings that jointly encode temporal dynamics and cross-series relational structure. We validated the effectiveness of our method on two downstream tasks. In GICS sector classification, our embeddings outperform benchmark approaches across macro, micro, and weighted F1 scores, demonstrating that they preserve economically meaningful structure. In the portfolio construction task, we used contextual similarity baskets in rolling backtests and observed consistent improvements over industry- and correlation-based baselines in terms of multiple risk-adjusted metrics. While validated on financial data, the proposed approach generalizes to domains where context-dependent temporal patterns matter—e.g., clinical signals, climate dynamics, or neural activity—demonstrating its flexibility and broad applicability.

Acknowledgments

The views expressed here are those of the authors alone and not of BlackRock, Inc.

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